Problem 5.7

Fund A calculates interest using exact simple interest (actual/actual). Fund B calculates interest using ordinary simple interest (30/360). Fund C calculates interest using the Banker's Rule (actual/360). All funds earn 5% simple interest and have the same amount of money deposited on January 1, 2005.

Order the funds based on the amount in the funds on March 1, 2005 from smallest to largest.

Let the principal be P.

Fund A: Exact Simple Interest

From January 1, 2005 to March 1, 2005, the exact number of days is

$$30 + 28 + 1 = 59 \text{ days.}$$

Thus,

$$I_A = P(0.05)\left(\frac{59}{365}\right).$$

Fund B: Ordinary Simple Interest

Using 30-day months,

$$30 + 30 + 1 = 61 \text{ days.}$$

So,

$$I_B = P(0.05)\left(\frac{61}{360}\right).$$

Fund C: Banker's Rule

Using the exact number of days with a 360-day year,

$$I_C = P(0.05)\left(\frac{59}{360}\right).$$

Comparison

Compare the time factors:

$$\frac{59}{365} < \frac{59}{360} < \frac{61}{360}.$$

Multiplying by the same positive quantity $0.05P$, we get

$$I_A < I_C < I_B.$$

Since all three funds start with the same principal, the final amounts satisfy the same order.

Therefore, from smallest to largest:

$$\boxed{A,\ C,\ B.}$$